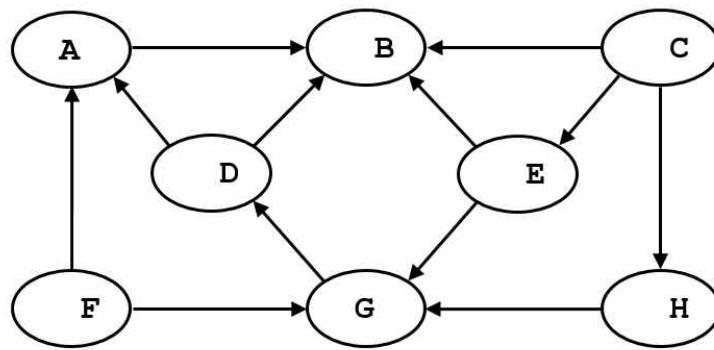


Algorithm Analysis Homework 5

Due by 5/29(Fri.) through LMS

Write a C++ program that implements topological sorting to find the linear ordering of a given directed acyclic graph (DAG). The DAG's structure is illustrated in the figure below as an example.



Your program should:

1. Graph Construction:

- In the main function, build an adjacency representation of the graph by adding vertices and edges using functions analogous to 'addVertex()' and 'addEdge()' (as discussed in class).
- The adjacency list must store neighbors of each vertex in alphabetical order.

2. Depth-First Search (DFS):

- Perform a DFS traversal starting from vertex 'A'.
- Calculate and display the start and finish times for each vertex.

3. Topological Sort:

- Apply topological sorting to determine the linear ordering of the vertices.
- Present the resulting topological ordering in a clear and readable format.

[Constraints]

- The graph will have a maximum of 10 nodes.

[Implementation Requirements]

1. Ensure the output is well-formatted and easy to understand.
2. You may utilize the Standard Template Library (STL) and other appropriate C++ features if necessary.
3. The code must compile and run successfully with g++.
4. Thoroughly test your program with the provided example and a few other additional DAGs.
4. Refrain from using large-scale AI code generation tools in the development of this program.
5. Include a header comment at the beginning of your code that lists all resources consulted during development. Follow the example format below:

```
// Lecture slide chapter 16.
```

```
// Blog: ** URL here **
```

```
// Book: "Algorithm analysis in C++" by [Author's Name]
```